18.05.2020. K1

1. Сигнал $f(t) = 0.2\cos(2\pi t) + 0.2\cos(3.6\pi t) + 0.2\cos(2.8\pi t)$ доводи се на улаз идеалног NF филтра граничне учестаности $\omega_c = 10\left[\frac{rad}{s}\right]$.

a) Одредити фреквенцију одмјеравања која гарантује реконструкцију филтрираног сигнала без губитака.

b) Ако се усвоји да је сигнал на излазу идеалног NF филтра означен са $y(t)$, написати израз за добијену поворку одбирака $y^*(t)$.

c) Скицирати одмјерени сигнал $y^*(t)$.

d) Наћи Z комплексни лик поворке одбирака $y^*(t)$.

РЈЕШЕЊЕ:

a) - У $f(t)$ постоје компоненте фреквенција

$2\pi = 6.28\left[\frac{rad}{s}\right]$

$3.6\pi = 11.31\left[\frac{rad}{s}\right] > \omega_c$

$2.8\pi = 8.80\left[\frac{rad}{s}\right]$

- Филтар не пропушта другу компоненту сигнала

- Након филтрације: $\omega_g = 8.80\left[\frac{rad}{s}\right]$

$\omega \geq 2\omega_g \Rightarrow \omega_s \geq 17.60\left[\frac{rad}{s}\right]$

$f_s = \frac{\omega_s}{2\pi} = \frac{17.60}{2\pi} \Rightarrow f_s \geq 2.8\,[Hz] \Rightarrow$ усвојено $f_s = 4\,[Hz]$

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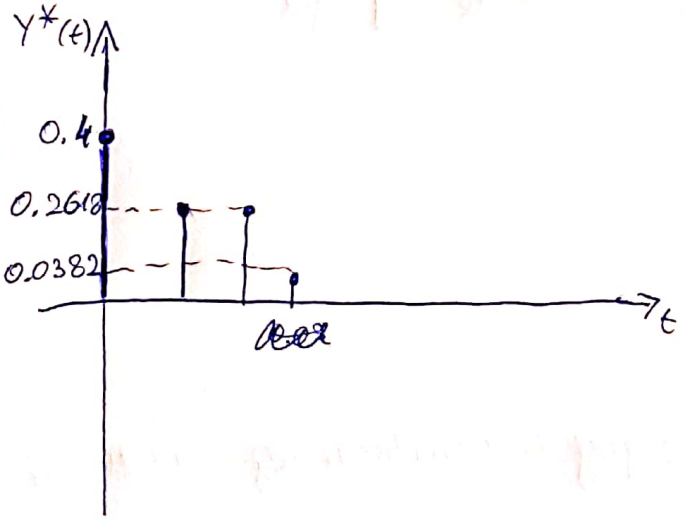
$$T_s = \frac{1}{f_s} = \frac{1}{4} \Rightarrow T_s = 0.25 \ [s]$$

b) $$y^*(t) = \sum_{k=0}^{\infty} y(t)\,\delta(t - kT_s) =$$

$$= \sum_{k=0}^{\infty} \left[0.2\cos(2\pi t) + 0.2\cos(2.8\pi t)\right]\delta(t - 0.25k)$$

c)



$$Y(0) = 0.2\cos(2\pi \cdot 0) + 0.2\cos(2.8\pi \cdot 0) = 0.4$$

$$Y(1) = 0.2\cos(4\pi) + 0.2\cos(5.6\pi) = 0.2618$$

$$Y(2) = 0.2\cos(6\pi) + 0.2\cos(8.4\pi) = 0.2618$$

$$Y(3) = 0.2\cos(8\pi) + 0.2\cos(11.2\pi) = 0.0382$$

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d) $Y(z) = Z\{0.2\cos(2\pi t) + 0.2\cos(2.8\pi t)\} =$

$$= 0.2\, Z\left\{\frac{e^{j2\pi t} + e^{-j2\pi t}}{2} + \frac{e^{j2.8\pi t} + e^{-j2.8\pi t}}{2}\right\} =$$

$$= 0.1\, Z\left\{e^{j2\pi t} + e^{-j2\pi t} + e^{j2.8\pi t} + e^{-j2.8\pi t}\right\}$$

$$F(z) = \sum_{k=0}^{+\infty} f(kT)\, z^{-k}$$

$$Y(z) = 0.1\left[\sum_{k=0}^{+\infty} e^{j2\pi kT} z^{-k} + \sum_{k=0}^{+\infty} e^{-j2\pi kT} z^{-k} + \right.$$

$$\left. + \sum_{k=0}^{+\infty} e^{j2.8\pi kT} z^{-k} + \sum_{k=0}^{+\infty} e^{-j2.8\pi kT} z^{-k}\right] =$$

$$= 0.1\left[\sum_{k=0}^{+\infty}\left(\frac{e^{j2\pi T}}{z}\right)^k + \sum_{k=0}^{+\infty}\left(\frac{e^{-j2\pi T}}{z}\right)^k + \right.$$

$$\left. + \sum_{k=0}^{+\infty}\left(\frac{e^{j2.8\pi T}}{z}\right)^k + \sum_{k=0}^{+\infty}\left(\frac{e^{-j2.8\pi T}}{z}\right)^k\right] =$$

$$= 0.1\left[\frac{1}{1 - \frac{e^{j2\pi T}}{z}} + \frac{1}{1 - \frac{e^{-j2\pi T}}{z}} + \right.$$

$$\left. + \frac{1}{1 - \frac{e^{j2.8\pi T}}{z}} + \frac{1}{1 - \frac{e^{-j2.8\pi T}}{z}}\right]$$

$$Y(z) = 0.1\left[\frac{z}{z - e^{j2\pi T}} + \frac{z}{z - e^{-j2\pi T}} + \frac{z}{z - e^{j2.8\pi T}} + \frac{z}{z - e^{-j2.8\pi T}}\right]$$

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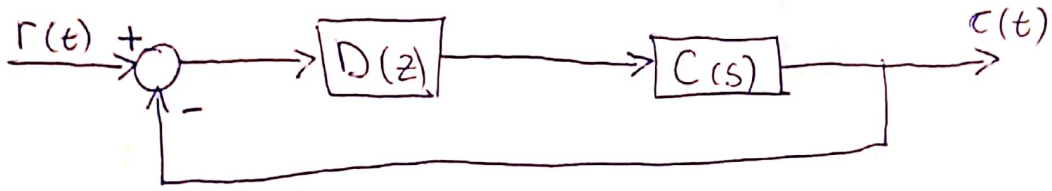
2. Непрекидан систем аутоматског управљања дат је на слици.

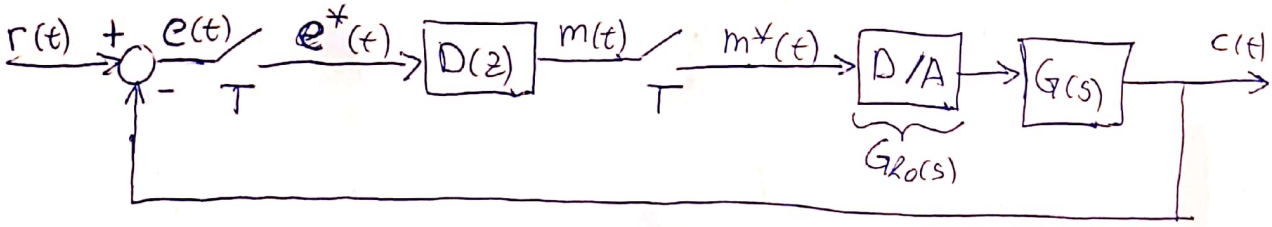


a) Нацртати систем са елементима који недостају.

b) Ако се усвоји да D/A конвертор ради као кола задршке нултог реда, да је D(z) представљен ј-ном $m(kT) = e(kT)$ и да је $G(s) = \frac{1}{s(s+2)}$, одредити вриједност сигнала $c(t)$ у тренутку $t = 1.3T$ и $t = 2.3T$ ако се на улаз доведе јединични одскочни сигнал. Усвојити да је периода одмјеравања $T = 1[s]$.

РЈЕШЕЊЕ:

a)



b) - Када је потребно одредити вриједност сигнала између тренутака одабирања користи се модификована Z трансформација (Z_m)

$$Z_m^{-1}\{F(z,m)\} = f[kT - (1-m)T]$$

- Нама треба: $t = 1.3T$ и $t = 2.3T \Rightarrow m = 0.3$

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$$C(s) = G_{h_0}(s) \cdot G(s) \cdot M^*(s) \xrightarrow[\text{ТАФДП}]{} C^*(s) = G_{h_0} G^*(s) \cdot M^*(s)$$

$$M(s) = D(z) \cdot E^*(s) \xrightarrow[\text{ТАФДП}]{} M^*(s) = D(z) \cdot E^*(s)$$

$$E(s) = R(s) - C(s) \xrightarrow[\text{ТАФДП}]{} E^*(s) = R^*(s) - C^*(s)$$

$$M^*(s) = D(z) \cdot E^*(s) = D(z) \cdot R^*(s) - D(z) \cdot C^*(s)$$

$$C^*(s) = G_{h_0} G^*(s) \cdot M^*(s) = G_{h_0} G^*(s) \cdot D(z) \cdot R^*(s) - G_{h_0} G^*(s) \cdot D(z) \cdot C^*(s)$$

$$C^*(s) = \frac{G_{h_0} G^*(s) \cdot D(z)}{1 + G_{h_0} G^*(s) \cdot D(z)} \cdot R^*(s) \Big/ \cdot e^{-(1-m)sT}$$ - зер нам треба z_m

$$\left[C(s) \cdot e^{-(1-m)sT}\right]^* = \frac{\left[G_{h_0} G(s) \cdot e^{-(1-m)sT}\right]^* \cdot D(z)}{1 + G_{h_0} G^*(s) \cdot D(z)} \cdot R^*(s) \Big/ Z$$

$$C(z,m) = \frac{G_{h_0} G(z,m) \cdot D(z)}{1 + G_{h_0} G(z) \cdot D(z)} \cdot R(z)$$

$$D(z): \; m(kT) = e(kT) \Big/ Z \Rightarrow \left.\begin{array}{l} M(z) = E(z) \\ M(z) = D(z) \cdot E(z) \end{array}\right\} \Rightarrow \underline{\underline{D(z) = 1}}$$

$$r(t) = 1(t) \Big/ Z \Rightarrow \underline{\underline{R(z) = \frac{z}{z-1}}}$$

$$G_{h_0} G(z) = Z\left\{\frac{1 - e^{-sT}}{s} \cdot \frac{1}{s(s+2)}\right\} = (1 - z^{-1}) Z\left\{\frac{1}{s^2(s+2)}\right\}$$

$$\frac{1}{s^2(s+2)} = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{s+2} = \frac{As + 2A + Bs^2 + 2Bs + Cs^2}{s^2(s+2)}$$

$$\left.\begin{array}{l} B + C = 0 \\ A + 2B = 0 \\ 2A = 1 \end{array}\right\} \Rightarrow A = \frac{1}{2}, \; B = -\frac{1}{4}, \; C = \frac{1}{4}$$

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$$G_{h_0}G(z) = \frac{1}{4}(1-z^{-1})\,Z\left\{\frac{2}{s^2} - \frac{1}{s} + \frac{1}{s+2}\right\}$$

$$G_{h_0}G(z) = \frac{1}{4}\cdot\frac{z-1}{z}\left[\frac{2Tz}{(z-1)^2} - \frac{z}{z-1} + \frac{z}{z-e^{-2T}}\right], \quad T = 1[s]$$

$$G_{h_0}G(z) = \frac{1}{4}\cdot\frac{z-1}{z}\left[\frac{2z}{(z-1)^2} - \frac{z}{z-1} + \frac{z-1}{z-0.1353}\right]$$

$$G_{h_0}G(z) = \frac{1}{4}\left[\frac{2}{z-1} - 1 + \frac{z-1}{z-0.1353}\right]$$

$$G_{h_0}G(z) = \frac{1}{4}\cdot\frac{2(z-0.1353) - (z-1)(z-0.1353) + (z-1)^2}{(z-1)(z-0.1353)}$$

$$G_{h_0}G(z) = \frac{1}{4}\cdot\frac{\cancel{2z} - 0.2706 - \cancel{z^2} + 1.1353z - 0.1353 + \cancel{z^2} - \cancel{2z} + 1}{(z-1)(z-0.1353)}$$

$$G_{h_0}G(z) = \frac{1}{4}\cdot\frac{1.1353z + 0.5941}{(z-1)(z-0.1353)} \Rightarrow \underline{\underline{G_{h_0}G(z) = \frac{0.2838z + 0.1485}{(z-1)(z-0.1353)}}}$$

$$G_{h_0}G(z,m) = Z_m\left\{\frac{1-e^{-sT}}{s}\cdot\frac{1}{s(s+2)}\right\} = (1-z^{-1})\,Z_m\left\{\frac{1}{s^2(s+2)}\right\}$$

$$G_{h_0}G(z,m) = \frac{1}{4}\cdot\frac{z-1}{z}\cdot Z_m\left\{\frac{2}{s^2} - \frac{1}{s} + \frac{1}{s+2}\right\}$$

$$G_{h_0}G(z,m) = \frac{1}{4}\cdot\frac{z-1}{z}\cdot\left[\frac{2mT}{z-1} + \frac{2T}{(z-1)^2} - \frac{1}{z-1} + \frac{e^{-2mT}}{z-e^{-2T}}\right]$$

$T = 1[s]$, $m = 0.3$

$$G_{h_0}G(z,m) = \frac{1}{4}\cdot\frac{z-1}{z}\left[\frac{0.6}{z-1} + \frac{2}{(z-1)^2} - \frac{1}{z-1} + \frac{e^{-0.6}}{z-e^{-2}}\right]$$

$$G_{h_0}G(z,m) = \frac{1}{4}\cdot\frac{z-1}{z}\left[\frac{2}{(z-1)^2} - \frac{0.4}{z-1} + \frac{0.5488}{z-0.1353}\right]$$

$$G_{h_0}G(z,m) = \frac{z-1}{z}\cdot\left[\frac{0.5}{(z-1)^2} - \frac{0.1}{z-1} + \frac{0.1372}{z-0.1353}\right]$$

$$G_{h_0}G(z,m) = \frac{\cancel{z-1}}{z}\cdot\frac{0.5(z-0.1353) - 0.1(z-1)(z-0.1353) + 0.1372(z-1)^2}{(z-1)^2(z-0.1353)}$$

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$$G_{h_0}G(z,m) = \frac{0.5z - 0.06765 - 0.1z^2 + 0.11353z - 0.01353 + 0.1372z^2 - 0.2744z + 0.1372}{z(z-1)(z-0.1353)}$$

$$G_{h_0}G(z,m) = \frac{0.0372z^2 + 0.33913z + 0.0562}{z(z-1)(z-0.1353)}$$

$$C(z,m) = \frac{\dfrac{0.0372z^2 + 0.33913z + 0.0562}{\cancel{z}(z-1)(z-0.1353)}}{1 + \dfrac{0.2838z + 0.1485}{(z-1)(z-0.1353)}} \cdot \frac{\cancel{z}}{z-1}$$

$$C(z,m) = \frac{\dfrac{0.0372z^2 + 0.33913z + 0.0562}{\cancel{(z-1)(z-0.1353)}}}{\dfrac{z^2 - 1.1353z + 0.1353 + 0.2838z + 0.1485}{\cancel{(z-1)(z-0.1353)}}} \cdot \frac{1}{z-1}$$

$$C(z,m) = \frac{0.0372z^2 + 0.33913z + 0.0562}{z^2 - 0.8515z + 0.2838} \cdot \frac{1}{z-1}$$

$$C(z,m) = \frac{0.0372z^2 + 0.33913z + 0.0562}{z^3 - z^2 - 0.8515z^2 + 0.8515z + 0.2838z - 0.2838}$$

$$C(z,m) = \frac{0.0372z^2 + 0.33913z + 0.0562}{z^3 - 1.8515z^2 + 1.1347z - 0.2838}$$

$$c[kT - (1-m)T] = Z_m^{-1}\{C(z,m)\}$$

$$(0.0372z^2 + 0.33913z + 0.0562) : (z^3 - 1.8515z^2 + 1.1347z - 0.2838) = 0.0372z^{-1} + 0.40803z^{-2} + 0.7695z^{-3} + \dots$$

$$-(0.0372z^2 - 0.0689z + 0.0422 - 0.0106)$$

$$0.40803z + 0.0140 + 0.0106z^{-1}$$

$$-(0.40803z - 0.7555 + 0.4630z^{-1} - 0.1158z^{-2})$$

$$0.7695 - 0.4524z^{-1} + 0.1158z^{-2}$$

$$\vdots$$

$$k=2 \Rightarrow c[2T - (1-0.3)T] = c[1.3T] = 0.40803$$

$$k=3 \Rightarrow c[3T - (1-0.3)T] = c[2.3T] = 0.7695$$

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③ Објекат аутоматског управљања описан је моделом у простору стања.

$$\dot{X}(t) = \begin{bmatrix} 0 & 1 \\ -15 & -8 \end{bmatrix} X(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$Y(t) = \begin{bmatrix} 10 & 0 \end{bmatrix} X(t)$$

а) Одредити модел објекта у дискретном простору стања.

б) Одредити ф-ју дискретног преноса објекта.

Усвојити да је периода одмјеравања $T = 0.2$ [s].

РЈЕШЕЊЕ:

а) $X(k+1) = E\,x(k) + C\,u(k)$

$Y(k) = C\,x(k)$

$$E(T) = e^{AT} = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}\Big|_{t=T} =$$

$$= \mathcal{L}^{-1}\left\{\left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -15 & -8 \end{bmatrix}\right)^{-1}\right\}\Big|_{t=T} = \mathcal{L}^{-1}\left\{\begin{bmatrix} s & -1 \\ 15 & s+8 \end{bmatrix}\right\}\Big|_{t=T} =$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{\underbrace{s^2+8s+15}_{(s+3)(s+5)}}\begin{bmatrix} s+8 & 1 \\ -15 & s \end{bmatrix}\right\}\Big|_{t=T} =$$

$$= \mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{s+8}{(s+3)(s+5)} & \frac{1}{(s+3)(s+5)} \\ \frac{-15}{(s+3)(s+5)} & \frac{s}{(s+3)(s+5)} \end{bmatrix}\right\}\Big|_{t=T}$$

$$\frac{s+8}{(s+3)(s+5)} = \frac{A}{s+3} + \frac{B}{s+5} = \frac{As+5A+Bs+3B}{(s+3)(s+5)}$$

$$\left.\begin{aligned} A+B &= 1 \\ 5A+3B &= 8 \end{aligned}\right\} \Rightarrow \begin{aligned} -3A-3B &= -3 \\ 5A+3B &= 8 \end{aligned} \Rightarrow 2A = 5 \Rightarrow A = \frac{5}{2},\ B = -\frac{3}{2}$$

$$\frac{1}{(s+3)(s+5)} = \frac{A}{s+3} + \frac{B}{s+5} = \frac{As+5A+Bs+3B}{(s+3)(s+5)}$$

$$\left.\begin{aligned} A+B &= 0 \\ 5A+3B &= 1 \end{aligned}\right\} \Rightarrow A = \frac{1}{2},\ B = -\frac{1}{2}$$

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$$\frac{-15}{(s+3)(s+5)} = \frac{A}{s+3} + \frac{B}{s+5} = \frac{As+5A+Bs+3B}{(s+3)(s+5)}$$

$$\left.\begin{array}{l} A+B=0 \\ 5A+3B=-15 \end{array}\right\} \Rightarrow 2A=-15 \Rightarrow A=-\frac{15}{2},\ B=\frac{15}{2}$$

$$\frac{s}{(s+3)(s+5)} = \frac{A}{s+3} + \frac{B}{s+5} = \frac{As+5A+Bs+3B}{(s+3)(s+5)}$$

$$\left.\begin{array}{l} A+B=1 \\ 5A+3B=0 \end{array}\right\} \Rightarrow \left.\begin{array}{l} -3A-3B=-3 \\ 5A+3B=0 \end{array}\right\} \Rightarrow 2A=-3 \Rightarrow A=-\frac{3}{2}$$

$$B=\frac{5}{2}$$

$$E(T) = \mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{5/2}{s+3} - \frac{3/2}{s+5} & \frac{1/2}{s+3} - \frac{1/2}{s+5} \\ \frac{-15/2}{s+3} + \frac{15/2}{s+5} & \frac{-3/2}{s+3} + \frac{5/2}{s+5} \end{bmatrix}\right\}\Bigg|_{t=T} =$$

$$= \frac{1}{2}\mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{5}{s+3} - \frac{3}{s+5} & \frac{1}{s+3} - \frac{1}{s+5} \\ \frac{-15}{s+3} + \frac{15}{s+5} & \frac{-3}{s+3} + \frac{5}{s+5} \end{bmatrix}\right\}\Bigg|_{t=T} =$$

$$= \frac{1}{2}\begin{bmatrix} 5e^{-3t} - 3e^{-5t} & e^{-3t} - e^{-5t} \\ 15(-e^{-3t} + e^{-5t}) & -3e^{-3t} + 5e^{-5t} \end{bmatrix}\Bigg|_{t=T},\ T=0.2[s]$$

$$E(T) = \frac{1}{2}\begin{bmatrix} 5e^{-3\cdot 0.2} - 3e^{-5\cdot 0.2} & e^{-3\cdot 0.2} - e^{-5\cdot 0.2} \\ 15(-e^{-3\cdot 0.2} + e^{-5\cdot 0.2}) & -3e^{-3\cdot 0.2} - 5e^{-5\cdot 0.2} \end{bmatrix}$$

$$E(T) = \frac{1}{2}\begin{bmatrix} 1.6404 & 0.1809 \\ -2.7140 & 0.1930 \end{bmatrix} \Rightarrow \underline{\underline{E(T) = \begin{bmatrix} 0.8202 & 0.0907 \\ -1.3570 & 0.0965 \end{bmatrix}}}$$

$$F(T) = \int_0^T e^{At} B\,dt = \frac{1}{2}\int_0^T \begin{bmatrix} 5e^{-3t} - 3e^{-5t} & e^{-3t} - e^{-5t} \\ 15(-e^{-3t} + e^{-5t}) & -3e^{-3t} + 5e^{-5t} \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} dt$$

$$F(T) = \frac{1}{2}\int_0^T \begin{bmatrix} e^{-3t} - e^{-5t} \\ -3e^{-3t} + 5e^{-5t} \end{bmatrix} dt = \frac{1}{2}\begin{bmatrix} -\frac{1}{3}e^{-3t} + \frac{1}{5}e^{-5t} \\ e^{-3t} - 5e^{-5t} \end{bmatrix}\Bigg|_0^T$$

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$$F(T) = \frac{1}{2}\begin{bmatrix} -\frac{1}{3}e^{-3\cdot 0.2} + \frac{1}{5}e^{-5\cdot 0.2} + \frac{1}{3}e^{-3\cdot 0} - \frac{1}{5}e^{-3\cdot 0} \\ e^{-3\cdot 0.2} - e^{-5\cdot 0.2} - e^{-3\cdot 0} + e^{-5\cdot 0} \end{bmatrix}$$

$$F(T) = \frac{1}{2}\begin{bmatrix} 0.0240 \\ 0.1809 \end{bmatrix} \Rightarrow \underline{\underline{F(T) = \begin{bmatrix} 0.0120 \\ 0.0905 \end{bmatrix}}}$$

$$\boxed{\begin{aligned} X(k+1) &= E\,x(k) + F\,u(k) = \begin{bmatrix} 0.8202 & 0.0907 \\ -1.3570 & 0.0965 \end{bmatrix} x(k) + \begin{bmatrix} 0.0120 \\ 0.0905 \end{bmatrix} u(k) \\ Y(k) &= C\,x(k) = \begin{bmatrix} 10 & 0 \end{bmatrix} x(k) \end{aligned}}$$

дискретан модел система у простору стања

b) $X(k+1) = E\,x(k) + F\,u(k) \;/Z$

$z\,X(z) = E\,X(z) + F\,U(z)$

$(zI - E)X(z) = F\,U(z)$

$X(z) = (zI - E)^{-1} F\,U(z)$

$Y(k) = C\,x(k) \;/Z$

$Y(z) = C\,X(z)$

$$Y(z) = C\,X(z) = C(zI - E)^{-1} F\,U(z)$$

$$G(z) = \frac{Y(z)}{U(z)} = C(zI - E)^{-1} F$$ – ф-ја дискретног преноса система

$$G(z) = \begin{bmatrix} 10 & 0 \end{bmatrix}\left(\begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} - \begin{bmatrix} 0.8202 & 0.0907 \\ -1.3570 & 0.0965 \end{bmatrix}\right)^{-1} \begin{bmatrix} 0.0120 \\ 0.0905 \end{bmatrix}$$

$$G(z) = \begin{bmatrix} 10 & 0 \end{bmatrix} \begin{bmatrix} z - 0.8202 & -0.0907 \\ 1.3570 & z - 0.0965 \end{bmatrix}^{-1} \begin{bmatrix} 0.0120 \\ 0.0905 \end{bmatrix}$$

$$G(z) = \begin{bmatrix} 10 & 0 \end{bmatrix} \cdot \frac{1}{(z - 0.8202)(z - 0.0965) + 0.1231} \begin{bmatrix} z - 0.0965 & 0.0907 \\ -1.3570 & z - 0.8202 \end{bmatrix} \cdot \begin{bmatrix} 0.0120 \\ 0.0905 \end{bmatrix}$$

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$$G(z) = \frac{1}{z^2 - 0.9167z + 0.2023} \begin{bmatrix} 10z - 0.9650 & 0.9070 \end{bmatrix} \begin{bmatrix} 0.0120 \\ 0.0905 \end{bmatrix}$$

$$G(z) = \frac{0.1200z - 0.01158 + 0.0821}{z^2 - 0.9167z + 0.2023} \Rightarrow \boxed{G(z) = \frac{0.12z + 0.0705}{z^2 - 0.9167z + 0.2023}}$$

ф-ja дискретной переноса система

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